Type MS · Quadratic trig equations using an identity

Answer 1 of 11 · Question 10 part c · 4 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

Number		
10 (a)	$x = \tan^{-1}(-3) = -72$	M1
	$x = 108 \qquad x = 288$	A1 A1
		(3)
(b)	$7\sin^2\theta + \sin\theta\cos\theta = 6(\sin^2\theta + \cos^2\theta)$	M1
	$\sin^2\theta + \sin\theta\cos\theta - 6\cos^2\theta = 0$	
	$\frac{\sin^2\theta}{\cos^2\theta} + \frac{\sin\theta}{\cos\theta} - 6 = 0$	M1
	$\tan^2\theta + \tan\theta - 6 = 0$	A1 cso
		(3)
(c)	$(\tan y + 3)(\tan y - 2) = 0$	M1
	$\tan y = -3 \qquad \tan y = 2$	A1
	$y = 108, 288 \qquad y = 63, 243$	A1ft A1
		(4)
		Total 10 marks

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Part	Mark	Guidance
(a)	M1	For using inverse tan to obtain any correct angle $\tan^{-1}(-3) \Rightarrow x = -71.565...^{o}$ Accept awrt -72°
	A1	For **either** 108 **or** 288
	A1	For **both** 108 **and** 288
(b)	M1	Uses $\sin^2\theta + \cos^2\theta = 1$ on the given equation to obtain $7\sin^2\theta + \sin\theta\cos\theta = 6(\sin^2\theta + \cos^2\theta)$
	M1	For rearranging and dividing through by $\cos^2\theta$ with the $\frac{\sin\theta}{\cos\theta} = \tan\theta$ identity to obtain a 3TQ: $\frac{\sin^2\theta}{\cos^2\theta} + \frac{\sin\theta}{\cos\theta} - 6 = 0 \Rightarrow \left(\tan^2\theta + \tan\theta - 6 = 0\right)$
	ALT 1	
	M1	Divides the given equation through by $\cos^2\theta$ with the $\frac{\sin\theta}{\cos\theta} = \tan\theta$ identity to obtain $7\tan^2\theta + \tan\theta = \frac{6}{\cos^2\theta}$
	M1	Uses $\sin^2\theta + \cos^2\theta = 1$ to obtain $\tan^2\theta + 1 = \frac{1}{\cos^2\theta}$ and uses this result on the given equation and rearranges to achieve a 3TQ to obtain $7\tan^2\theta + \tan\theta = 6\left(1 + \tan^2\theta\right) \Rightarrow \left(\tan^2\theta + \tan\theta - 6 = 0\right)$
	A1	For obtaining the given expression $\tan^2\theta + \tan\theta - 6 = 0$ * in full. **Note: This is a show question, there must be no errors in the solution.**
	ALT 2	
	M1	Uses the identity $\frac{\sin\theta}{\cos\theta} = \tan\theta$ with $\tan^2\theta + \tan\theta - 6 = 0$ to achieve $\frac{\sin^2\theta}{\cos^2\theta} + \frac{\sin\theta}{\cos\theta} - 6 = 0$ and multiplies through by $\cos^2\theta$ to obtain $\sin^2\theta + \sin\theta\cos\theta - 6\cos^2\theta = 0$
	M1	Uses $\sin^2\theta + \cos^2\theta = 1$ to obtain $\sin^2\theta + \sin\theta\cos\theta - 6\left(1 - \sin^2\theta\right) = 0$ and rearranges to obtain $7\sin^2\theta + \sin\theta\cos\theta = 6$
	A1 cso	For obtaining the given expression $7\sin^2\theta + \sin\theta\cos\theta = 6$ in full. **Note: This is a show question, there must be no errors in the solution.**
(c)	M1	For changing $7\sin^2 y + \sin y\cos y = 6$ to $\tan^2 y + \tan y - 6 = 0$ [this step must be correct] and then attempting to solve the 3TQ by any method.
	A1	For $\tan y = -3$ and $\tan y = 2$
	A1ft	For **both** $y = 108$ **and** 288 (ft from (a)) **Do not ft angles out of range**
	A1	For **both** $y = 63$ **and** 243
Rounding errors: Penalise rounding only once in this question when first seen provided angles round to 108, 288, 63 or 243		
Extra angles: Deduct one A mark for any extra angles within range. Ignore angles outside of range.		

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Question number	Scheme	Marks

Type MS · Quadratic trig equations using an identity

Answer 2 of 11 · Question 11 part b · 4 marks

2021 JUN · Paper 01 · extract 1 of 5 · find part b in this mark scheme extract

Number		
11 (a)	$\cos 2x = \cos(x+x)$	
	$= \cos^2 x - \sin^2 x \ = 1 - \sin^2 x - \sin^2 x$	M1 M1
	$= 1 - 2\sin^2 x$ *	A1 cso (3)
(b)	$\sin x + 2\sin^2 x - 1 = 0 \Rightarrow \ (2\sin x - 1)(\sin x + 1) = 0$	M1dM1
	When $\sin x = \frac{1}{2}$ When $\sin x = -1$	
	$x = \frac{\pi}{6}, \frac{5\pi}{6}$ $x = \frac{3\pi}{2}$	A1
	$\left(\frac{\pi}{6}, \frac{5}{2}\right), \left(\frac{5\pi}{6}, \frac{5}{2}\right), \left(\frac{3\pi}{2}, 1\right)$	A1 (4)
(c)	$\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (\sin x + 2)\,dx - \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (\cos 2x + 2)\,dx$	M1
	$\left[-\cos x + 2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}} - \left[\frac{\sin 2x}{2} + 2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$	A1
	$\left[\left(-\cos\frac{5\pi}{6} + 2\left(\frac{5\pi}{6}\right)\right) - \left(-\cos\frac{\pi}{6} + 2\left(\frac{\pi}{6}\right)\right)\right]$	
	$-\left[\left(\frac{\sin 2\left(\frac{5\pi}{6}\right)}{2} + 2\left(\frac{5\pi}{6}\right)\right) - \left(\frac{\sin 2\left(\frac{\pi}{6}\right)}{2} + 2\left(\frac{\pi}{6}\right)\right)\right]$	M1
	$= \frac{3\sqrt{3}}{2}$	A1
	$\int_{\frac{5\pi}{6}}^{\frac{3\pi}{2}} (\cos 2x + 2)\,dx - \int_{\frac{5\pi}{6}}^{\frac{3\pi}{2}} (\sin x + 2)\,dx$	M1
	$\left[\frac{\sin 2x}{2} + 2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}} - \left[-\cos x + 2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}$	
	$\left[\left(\frac{\sin 2\left(\frac{3\pi}{2}\right)}{2} + 2\left(\frac{3\pi}{2}\right)\right) - \left(\frac{\sin 2\left(\frac{5\pi}{6}\right)}{2} + 2\left(\frac{5\pi}{6}\right)\right)\right]$	M1
	$-\left[\left(-\cos\left(\frac{3\pi}{2}\right) + 2\left(\frac{3\pi}{2}\right)\right) - \left(-\cos\left(\frac{5\pi}{6}\right) + 2\left(\frac{5\pi}{6}\right)\right)\right]$	
	$\frac{3\sqrt{3}}{4}$	A1
	$R_1 : R_2 = 2 : 1$	B1ft
	See Special Case at end of notes	(8)
		Total 15 marks

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Answer 2 of 11 · Question 11 part b · 4 marks

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Part	Mark	Notes
(a)	M1	For using the addition formula on page 2 and reaching the result $\cos 2x = \cos^2 x - \sin^2 x$ This **must** be correct for this mark.
	M1	For using **and** applying $\sin^2 x + \cos^2 x = 1$ in their $\cos 2x$ $\cos 2x = 1 - \sin^2 x - \sin^2 x$
	A1 cso	For the correct identity as shown in full. $\cos 2x = 1 - 2\sin^2 x$ * **Note: This is a show question, there must be no errors seen.**
(b)	M1	Equates the two given equations together. $\sin x + 2 = \cos 2x + 2$ and uses the given identity in part (a) to attempt to form a 3TQ in sin x: $\sin x + 2\sin^2 x - 1 = 0$ or $2\sin^2 x + \sin x - 1 = 0$ At least either $2\sin^2 x$ or $\sin x$ must be correct.
	dM1	Solves their 3TQ by any method to reach two values for $\sin x$ -see General Guidance $(2\sin x - 1)(\sin x + 1) = 0 \Rightarrow \sin x = \left(-1,\ \frac{1}{2}\right)$ This mark is dependent on the previous M mark.
	A1	For finding all three values of x: $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{3\pi}{2}$ Allow $x = 30^\circ,\ 150^\circ,\ 270^\circ$ for this mark – condone missing degree signs
	A1	For all three sets of correct coordinates: $\left(\frac{\pi}{6}, \frac{5}{2}\right), \left(\frac{5\pi}{6}, \frac{5}{2}\right), \left(\frac{3\pi}{2}, 1\right)$ Allow $\left(30^\circ, \frac{5}{2}\right), \left(150^\circ, \frac{5}{2}\right), \left(270^\circ, 1\right)$ – condone missing degree signs
(c)	**Area R_1**	
	M1	For stating a correct method to find the area R_1 using their limits from part (b) of $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ applied the correct way. Condone angles in degrees. $\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (\sin x + 2)\,dx - \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (\cos 2x + 2)\,dx$ (The + 2 – 2 may have been simplified).
	M1	For an attempt to integrate their expression for R_1 which must involve $\sin x$ and $\cos 2x$ A minimally acceptable integral of $\sin x \to -\cos x$ **or** $\cos 2x \to \pm\frac{\sin 2x}{2}$ and $2 \to 2x$ (The $2 \to 2x$ may not be present if they've simplified first). At least one trig term must be acceptable **with** $2 \to 2x$ (if present) correct. $\left[-\cos x + 2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}} - \left[\frac{\sin 2x}{2} + 2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$ Ignore incorrect, angles in degrees or even absent limits for this mark.

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Answer 2 of 11 · Question 11 part b · 4 marks

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	M1	Substitutes their limits correctly into their integrated expression. Must involve a changed expression. $\left[\left(-\cos\frac{5\pi}{6}+2\left(\frac{5\pi}{6}\right)\right)-\left(-\cos\frac{\pi}{6}+2\left(\frac{\pi}{6}\right)\right)\right]-$ $\left[\left(\frac{\sin 2\left(\frac{5\pi}{6}\right)}{2}+2\left(\frac{5\pi}{6}\right)\right)-\left(\frac{\sin 2\left(\frac{\pi}{6}\right)}{2}+2\left(\frac{\pi}{6}\right)\right)\right]$ **Do not accept a substitution in degrees – it must be in radians.** **If the work for the first two method marks to find the area of R_1 is fully correct, this mark can be implied by $\frac{3\sqrt{3}}{2}$. If the student does not get** $\left[-\cos x+2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}-\left[\frac{\sin 2x}{2}+2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$ **from the first two methods marks, they must show the clear substitution of their limits to be awarded this mark.**
	A1	For the correct area.
Area R_2		
	M1	For an attempt to integrate their expression for R_2 which must involve $\int(\cos 2x+2)\,dx-\int(\sin x+2)\,dx$ or $\int\cos 2x-\sin x\,dx$ if they have simplified the +2 – 2. Ignore incorrect, angles in degrees or even absent limits for this mark. A minimally acceptable integral of $\sin x\to-\cos x$ or $\cos 2x\to\pm\frac{\sin 2x}{2}$ and $2\to 2x$ (The $2\to 2x$ may not be present if they've simplified first). At least one trig term must be acceptable **with** $2\to 2x$ (if present) correct. $\left[\frac{\sin 2x}{2}+2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}-\left[-\cos x+2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}$
	M1	For substituting their limits correctly into their integrated expression. Must involve a changed expression. $\left[\left(\frac{\sin 2\left(\frac{3\pi}{2}\right)}{2}+2\left(\frac{3\pi}{2}\right)\right)-\left(\frac{\sin 2\left(\frac{5\pi}{6}\right)}{2}+2\left(\frac{5\pi}{6}\right)\right)\right]$ $-\left[\left(-\cos\left(\frac{3\pi}{2}\right)+2\left(\frac{3\pi}{2}\right)\right)-\left(-\cos\left(\frac{5\pi}{6}\right)+2\left(\frac{5\pi}{6}\right)\right)\right]$ **Do not accept a substitution in degrees – it must be in radians.** **If the student's work for the first method mark to find the area of R_2 is fully correct, this M mark can be implied by $\frac{3\sqrt{3}}{4}$. If the student does not get**

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		$\left[\frac{\sin 2x}{2}+2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}-\left[-\cos x+2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}$ **from the first method mark, they must show the clear substitution of their limits to be awarded this mark.**
	A1	For the correct area.
Ratio	**area of R_1: area of R_2**	
	B1ft	For a ratio given in its simplest form $a : b$ where a, b are positive integers, ft their values for the area of R_1 *and* R_2 Correct ratio is $R_1:R_2 = 2 : 1$
	SC	**Special case: For the use of degrees in part (c).** **<u>If the area of R_1 and R_2 are both correct</u> when limits in degrees have been substituted** (possible, for example, when the candidate has simplified before integrating): Penalise the first substitution in degrees for the area R_1 by awarding M1 M1 M0 A0. For the area R_2 allow M1 M1 A1 if the substitution (in degrees) and the area of R_2 is correct. B1ft for the correct ratio.

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Type MS · Quadratic trig equations using an identity

Answer 3 of 11 · Question 10 part b · 5 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

10 (a)(i)	$\cos(A+A)=\cos A\cos A-\sin A\sin A\rightarrow\cos 2A=\cos^2 A-\sin^2 A$ $\cos 2A=1-\sin^2 A-\sin^2 A\Rightarrow\cos 2A=1-2\sin^2 A$ * $\cos 2A-1=-2\sin^2 A$ or $1-\cos 2A=2\sin^2 A$ $\sin^2 A=\frac{1-\cos 2A}{2}$ *	M1 M1 A1*cso [3]
/ALT (a)(i)	Starting from r.h.s of equation: $\cos(A+A)=\cos A\cos A-\sin A\sin A\rightarrow\cos 2A=\cos^2 A-\sin^2 A$ Gives $\frac{1-\cos 2A}{2}=\frac{1-(\cos^2 A-\sin^2 A)}{2}$ $=\frac{\sin^2 A+\sin^2 A}{2}=\sin^2$ as required.	M1 M1 A1*cso [3]
(ii)	$\left(\cos 2A=\cos^2 A-\sin^2 A\right.$ or $\left.\cos 2A=1-2\sin^2 A\right)$ $\cos 2A=1-\sin^2 A-\sin^2 A$ or $\cos 2A=1-2(1-\cos^2 A)$ $\rightarrow\cos 2A=2\cos^2 A-1$ $\cos 2A+1=2\cos^2 A$ $\cos^2 A=\frac{\cos 2A+1}{2}$ *	M1 A1*cso [2]
ALT (ii)	Starting from r.h.s of equation: $\cos 2A=\cos^2 A-\sin^2 A$ Gives $\frac{\cos 2A+1}{2}=\frac{\cos^2 A-\sin^2 A+1}{2}$ $=\frac{\cos^2 A+\cos^2 A}{2}=\cos^2 A$ as required.	 M1 A1*cso [2]
(b)	$\cos 2x=\sin x$ $1-2\sin^2 x=\sin x\rightarrow 0=2\sin^2 x+\sin x-1$ $0=(2\sin x-1)(\sin x+1)$ $\sin x=\frac{1}{2},(-1)\rightarrow x=$ $\left(\frac{\pi}{6},\frac{1}{2}\right)$	M1 M1 M1 A1 A1 [5]

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(c)	$\pi\int_0^{\frac{\pi}{6}}\left(\cos^2 2x-\sin^2 x\right)dx$	B1ft
	$(\pi)\int_0^{\frac{\pi}{6}}\left(\frac{\cos 4x+1}{2}\right)-\left(\frac{1-\cos 2x}{2}\right)dx$	M1
	$\left[(\pi)\int_0^{\frac{\pi}{6}}\left(\frac{\cos 4x}{2}+\frac{\cos 2x}{2}\right)dx\right]$	
	$(\pi)\left[\frac{\sin 4x}{8}+\frac{\sin 2x}{4}\right]_0^{\frac{\pi}{6}}$	M1
	$(\pi)\left(\frac{\sin 4\left(\frac{\pi}{6}\right)}{8}+\frac{\sin 2\left(\frac{\pi}{6}\right)}{4}\right)$	M1
	$\frac{3\sqrt{3}}{16}\pi$	A1 [5]
		Total 15 marks

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Part	Mark	Additional Guidance
10 (a)(i)	M1	Correct use of the addition formula as shown leading to $\cos 2A = \cos^2 A - \sin^2 A$
	M1	Correct substitution of $1-\sin^2 A$ to get $\cos 2A = 1 - 2\sin^2 A$
	A1*cso	Fully correct solution with the minimum steps shown, no errors.
ALT (a)(i)	M1	Correct use of the addition formula as shown leading to $\cos 2A = \cos^2 A - \sin^2 A$
	M1	Correct substitution into the right hand side of the equation.
	A1*cso	Rearrangement of resultant expression to reach left hand side of original equation., without errors..
(ii)	M1	Replacement of $\sin^2 A$ with $1-\cos^2 A$ in an expression from their part (i)
	A1*cso	Rearrangement of a correct equation to attain the given result, minimum steps shown, no errors.
ALT (ii)	M1	Replacement pf cos2A with expression from (i)
	A1*cso	Rearrangement of resultant expression to reach left hand side of original equation, without errors.
(b)	M1	Correctly equates the 2 expressions
	M1	Correctly uses the identity for cos2x and rearranges to an equation of the form $0 = a\sin^2 x + b\sin x + c \qquad a,b,c \neq 0$
	M1	Uses a full method to solve their equation of the form $0 = a\sin^2 x + b\sin x + c \qquad a,b,c \neq 0$ to find the inverse sine to arrive at a value for x. See general guidance for the solution of a 3TQ.
	A1	For the correct value of x
	A1	For correct coordinates, allow omission of brackets or x = and y = . **SC**: Allow A1 where candidate works in degrees and gets x = 30 degrees
(c)	B1ft	For the integral shown, ft their limits only.
	M1	An attempt to replace $\cos^2 2x$ **and** $\sin^2 x$ with an identity from part (a). At least one replacement must be fully correct.
	M1	At least one term correctly integrated.
	M1	Substitution of their limits clearly seen into a changed expression. This can be implied by a correct answer. Where the expression is not correct, or their limits are not correct, the substitution must be seen into at least half of their terms.
	A1	For the correct answer. Withold this mark if the candidate works in degrees rather than radians

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Question number	Scheme	Marks

Type MS · Quadratic trig equations using an identity

Answer 4 of 11 · Question 11 part b · 7 marks

2025 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

11 (a)	$\cos^4\theta-\sin^4\theta=(\cos^2\theta-\sin^2\theta)(\cos^2\theta+\sin^2\theta)$	M1
	$=(\cos^2\theta-\sin^2\theta)\times1=\cos2\theta$ *	M1M1A1 cso [4
	ALT example $\cos^4\theta-\sin^4\theta=\cos^4\theta-\sin^2\theta\sin^2\theta$	
	$=\cos^4\theta-(1-\cos^2\theta)(1-\cos^2\theta)$	2nd M1
	$=\cos^4\theta-(1-2\cos^2\theta+\cos^4\theta)$	1st M1
	$=(1-2\cos^2\theta)$	3rd M1
	$=\cos2\theta$	A1
(b)	$8\cos^2\left(2\theta+\frac{\pi}{4}\right)-3=2[\cos^4\left(\theta+\frac{\pi}{8}\right)-\sin^4\left(\theta+\frac{\pi}{8}\right)]$	M1
	$=2\cos\left(2\theta+\frac{\pi}{4}\right)$	A1
	$8\cos^2\left(2\theta+\frac{\pi}{4}\right)-2\cos\left(2\theta+\frac{\pi}{4}\right)-3=0$	M1
	$\left(4\cos\left(2\theta+\frac{\pi}{4}\right)-3\right)\left(2\cos\left(2\theta+\frac{\pi}{4}\right)+1\right)[=0]$	
	$\cos\left(2\theta+\frac{\pi}{4}\right)=\frac{3}{4}\quad\cos\left(2\theta+\frac{\pi}{4}\right)=-\frac{1}{2}$	A1
	$2\theta+\frac{\pi}{4}=0.72273...\quad2\theta+\frac{\pi}{4}=\frac{2\pi}{3}$	dM1
	$\theta=2.39,3.11,\ 0.65,1.70$ accept $\theta=\frac{5\pi}{24},\frac{13\pi}{24}$	A1A1 [7]
(c)	$\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\left(\cos^4 2x-\sin^4 2x-8\sin4x\right)\mathrm{d}x=\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\left(\cos4x-8\sin4x\right)\mathrm{d}x$	M1
	$=\left[\frac{1}{4}\sin4x+\frac{8}{4}\cos4x\right]_{\frac{\pi}{16}}^{\frac{\pi}{8}}$	M1
	$=\frac{1}{4}\left[\sin\left(4\times\frac{\pi}{8}\right)-\sin\left(4\times\frac{\pi}{16}\right)\right]+2\left[\cos\left(4\times\frac{\pi}{8}\right)-\cos\left(4\times\frac{\pi}{16}\right)\right]$	M1
	$=\frac{1}{4}\left[1-\frac{\sqrt{2}}{2}\right]+2\left[0-\frac{\sqrt{2}}{2}\right]=\frac{1}{4}-\frac{9\sqrt{2}}{8}$	A1 [4]
		Total 15 marks

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Part	Mark	Additional Guidance
General Principles for marking part (a) • First M mark is for either applying the difference of two squares or multiplying out their expression and collecting terms. • Second M mark is for applying $\cos^2\theta+\sin^2\theta=1$ in any form. This can be implied from correct work. However, do not award for just $\cos^4\theta-\sin^4\theta=\cos^2\theta-\sin^2\theta$ seen. • Third M mark is for applying any identity for $\cos 2\theta$ • A1 For obtaining the required result with no errors.		
(a)	M1	For factorising as the difference of two squares $(\cos^2\theta-\sin^2\theta)(\cos^2\theta+\sin^2\theta)$
	M1	For use of $\cos^2\theta+\sin^2\theta)=1$
	M1	For use of $\cos^2\theta-\sin^2\theta=\cos(2\theta)$
	A1 cso	Obtain the given result with no errors in the working.
(b)	M1	For using the given result to obtain a quadratic in $\left(2\theta+\frac{\pi}{4}\right)$ Note: another variable may be used in place of $\left(2\theta+\frac{\pi}{4}\right)$ for this mark
	A1	For rearranging to the correct quadratic
	M1	For a valid attempt to solve their **3TQ**. See General Guidance for what constitutes an attempt to factorise. If candidates use a calculator [so no working shown] then award this mark only if the 3TQ is correct with the correct solutions.
	A1	For obtaining correct values for $\cos\left(2\theta+\frac{\pi}{4}\right)$ Note: another variable may be used in place of $\left(2\theta+\frac{\pi}{4}\right)$ for this mark
	dM1	For achieving any valid value **in radians** for their $2\theta+\frac{\pi}{4}$ i.e. 0.7227…or $\frac{2}{3}\pi$ Accept values out of range for this mark. This is dependent on the previous M mark.
	A1	For two correct solutions in range
	A1	For all four correct solutions in range and no others. Extra solutions within range scores A0 Ignore solutions outside of range. Exact solutions when given must be given as multiples of π.
(c)	M1	For use of the given result to obtain $\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\cos 4x\ \mathrm{d}x$
	M1	For an attempt to integrate $\cos kx\rightarrow l\sin kx$ where $k,l\neq 1$
	M1	For substituting limits into a changed expression
	A1	For the correct answer in the required form.

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11(a)	$\frac{1-\cos 2A}{1+\cos 2A}=\frac{1-\left(\cos^2 A-\sin^2 A\right)}{1+\left(\cos^2 A-\sin^2 A\right)}$	M1
	$=\frac{1-\left(1-\sin^2 A-\sin^2 A\right)}{1+\left(\cos^2 A-\left(1-\cos^2 A\right)\right)}$	M1
	$=\frac{2\sin^2 A}{2\cos^2 A}=\tan^2 A$ *	A1cso [3]
(b)	$\frac{3-3\cos 4x}{1+\cos 4x}+\frac{5\sin 2x}{\cos 2x}=2\Rightarrow 3\tan^2 2x+5\tan 2x-2=0$	M1,M1
	$\Rightarrow\left(3\tan 2x-1\right)\left(\tan 2x+2\right)=0\Rightarrow\tan 2x=\frac{1}{3},-2$	M1A1
	$\Rightarrow 2x=-161.565, 18.43, -63.43, 116.56$	M1
	$\Rightarrow x=-80.8,\ -31.7,\ 9.2,\ 58.3$	A1A1
	Accept awrt 1dp	[7]
		Total 10 marks

Type MS · Quadratic trig equations using an identity

Answer 6 of 11 · Question 11 part c · 4 marks

2024 JUN · Paper 02 · extract 1 of 4 · find part c in this mark scheme extract

11(a)(i)	$\cos 2A = \cos^2 A - \sin^2 A, = \cos^2 A - \left(1 - \cos^2 A\right)$ $\cos 2A = 2\cos^2 A - 1$ *	M1,M1 A1 cso [3]
(ii)	$\sin 2A = \sin A\cos A + \sin A\cos A = 2\sin A\cos A$ *	B1 cso [1]
(b)	$\cos 3A = \cos\left(2A + A\right)$ $= \cos 2A\cos A - \sin A\sin 2A$ $= \left(2\cos^2 A - 1\right)\cos A - 2\sin A\sin A\cos A$ $= 2\cos^3 A - \cos A - 2\left(1 - \cos^2 A\right)\cos A$ $= 2\cos^3 A - \cos A + 2\cos^3 A - 2\cos A$ $= 4\cos^3 A - 3\cos A$ $\Rightarrow \cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$ * **ALT** $\cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$ $= \dfrac{\cos 2A\cos A - \sin 2A\sin A + 3\cos A}{4}$ $= \dfrac{\left(2\cos^2 A - 1\right)\cos A - 2\sin A\sin A\cos A + 3\cos A}{4}$ $= \dfrac{2\cos^3 A - \cos A - 2\left(1 - \cos^2 A\right)\cos A + 3\cos A}{4}$ $= \dfrac{2\cos^3 A - \cos A + 2\cos^3 A - 2\cos A + 3\cos A}{4}$ $= \dfrac{4\cos^3 A}{4} = \cos^3 A \quad [LHS = RHS]$	M1 M1 M1 A1 Cso [4] [M1 M1 M1 A1]
(c)	$\left[\cos 3A = 4\cos^3 A - 3\cos A \Rightarrow 2\cos 3A = 8\cos^3 A - 6\cos A\right]$ $8\cos^3\left(\dfrac{\theta}{2}\right) - 6\cos\left(\dfrac{\theta}{2}\right) - 1 = 0 \Rightarrow 2\cos 3\left(\dfrac{\theta}{2}\right) - 1 = 0$ $\cos 3\left(\dfrac{\theta}{2}\right) = \dfrac{1}{2} \Rightarrow 3\left(\dfrac{\theta}{2}\right) = \dfrac{\pi}{3}, \dfrac{5\pi}{3}, \dfrac{7\pi}{3}, \left(\dfrac{11\pi}{3}\right)$ $\theta = \dfrac{2\pi}{9}, \dfrac{10\pi}{9}, \dfrac{14\pi}{9}$	M1 M1 M1A1 [4]

Type MS · Quadratic trig equations using an identity

Answer 6 of 11 · Question 11 part c · 4 marks

2024 JUN · Paper 02 · extract 2 of 4 · find part c in this mark scheme extract

(d)	$I = \int_0^{\frac{\pi}{6}} \left(4\cos^3\theta - \sin 2\theta\right) \mathrm{d}\theta = \int_0^{\frac{\pi}{6}} \left(\cos 3\theta + 3\cos\theta - \sin 2\theta\right) \mathrm{d}\theta$	M1
	$I = \left[\frac{\sin 3\theta}{3} + 3\sin\theta + \frac{\cos 2\theta}{2}\right]_0^{\frac{\pi}{6}}$	M1
	$= \left[\frac{\sin 3\left(\frac{\pi}{6}\right)}{3} + 3\sin\left(\frac{\pi}{6}\right) + \frac{\cos 2\left(\frac{\pi}{6}\right)}{2}\right] - \left[\frac{\sin 3(0)}{3} + 3\sin(0) + \frac{\cos 2(0)}{2}\right]$	M1
	$= \left[\frac{1}{3} + \frac{3}{2} + \frac{1}{4}\right] - \left[0 - 0 + \frac{1}{2}\right] = \frac{19}{12}$	A1 [4]
		Total 16 marks

Type MS · Quadratic trig equations using an identity

Answer 6 of 11 · Question 11 part c · 4 marks

2024 JUN · Paper 02 · extract 3 of 4 · find part c in this mark scheme extract

(c)	M1	Substitutes $2\cos 3\left(\frac{\theta}{2}\right)$ in place of $8\cos^3\left(\frac{\theta}{2}\right)-6\cos\left(\frac{\theta}{2}\right)$ correctly to obtain at least $k\cos 3\left(\frac{\theta}{2}\right)-1=0$ where k is a constant
	M1	Rearranges to obtain $\cos 3\left(\frac{\theta}{2}\right)=\frac{1}{k}$ and takes the inverse cosine to find at least any one **correct** value for $3\left(\frac{\theta}{2}\right)$
	M1	Finds at least one correct value for θ
	A1	Finds all three values with no extra values within range. Ignore any values given outside of the range.
(d)	M1	For a **correct** substitution as shown. $4\cos^3\theta=\cos 3\theta+3\cos\theta$
	M1	For an acceptable attempt to integrate either $\cos 3\theta\rightarrow\pm\frac{\sin 3\theta}{3}$ or $-\sin 2\theta\rightarrow\pm\frac{\cos 2\theta}{2}$
	M1	For substituting both given values the correct way around. This mark can be awarded for a complete substitution SEEN into any changed expression. Award this mark for **correct** integration **and** a value of $\frac{19}{12}$ seen without explicit substitution.
	A1	For $\frac{19}{12}$

Type MS · Quadratic trig equations using an identity

Answer 6 of 11 · Question 11 part c · 4 marks

2024 JUN · Paper 02 · extract 4 of 4 · find part c in this mark scheme extract

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Type MS · Quadratic trig equations using an identity

Answer 7 of 11 · Question 8 part b · 5 marks

2024 NOV · Paper 01 · find part b in this mark scheme extract

Number		
8 (i) (a)	$\left(\tan 2A = \tan(A+A) =\right)\dfrac{\tan A+\tan A}{1-\tan A\tan A}$ $=\dfrac{2\tan A}{1-\tan^2 A}$ *	M1 A1cso* [2]
(b)	$\tan A-\dfrac{2\tan A}{1-\tan^2 A}=0$ $\tan A(1-\tan^2 A)-2\tan A=0$ $\tan A-\tan^3 A-2\tan A=0$ $\tan^3 A+\tan A=0$ $\tan A(\tan^2 A+1)=0$ $\tan A=0 \qquad A=0, 180$ $\left(\tan^2 A=-1 \qquad \text{no solutions}\right)$	M1 M1 M1 A1, A1 [5]
ALT	$\tan A-\dfrac{2\tan A}{1-\tan^2 A}=0$ $\tan A\left(1-\dfrac{2}{1-\tan^2 A}\right)=0 \Rightarrow 1-\tan^2 A-2=0 \text{ or } 2=1-\tan^2 A$ $\tan^2 A=-1$ $\tan A=0 \qquad A=0, 180$ $\left(\tan^2 A=-1 \qquad \text{no solutions}\right)$	
(ii)	$\cos x\cos\dfrac{\pi}{6}+\sin x\sin\dfrac{\pi}{6}\left(=\sin x\right)$ $\sin x+\sqrt{3}\cos x=2\sin x \Rightarrow \sqrt{3}=\tan x$ $x=-\dfrac{2\pi}{3},\dfrac{\pi}{3},\dfrac{4\pi}{3}$	M1 dM1 A1 A1 [4]
		Total 11 marks

Type MS · Quadratic trig equations using an identity

Answer 8 of 11 · Question 7 part c · 4 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

7 (a)	$(3x - 15 =)$ 30 **or** 330 **or** 390	M1
	$x = 15$ $x = 115$ $x = 135$	M1 A1 A1 [4]
(b)	$3\frac{\sin y}{\cos y} + 4\sin y = 0 \rightarrow \sin y\left(\frac{3}{\cos y} + 4\right) = 0$	M1
	$\sin y = 0$ and $\cos y = -\frac{3}{4}$ $\rightarrow$ $y =$	A1ft
	$y = -180, 0$	A1
	$y = 138.6, -138.6$	A1
	ALT	
	$3\tan y + 4\tan y \cos y = 0 \rightarrow \tan y\,(3 + 4\cos y) = 0$	M1
	$\tan y = 0$ and $\cos y = -\frac{3}{4}$ $\rightarrow$ $y =$	A1ft
	$y = -180, 0$	A1
	$y = 138.6, -138.6$	A1 [4]
(c)	$\cos\theta = 3\,(1 - \cos^2\theta) - 1$ $\rightarrow$ $3\cos^2\theta + \cos\theta - 2 = 0$	M1
	$(3\cos\theta - 2)(\cos\theta + 1)$	M1
	$\theta = -180$	A1
	$\theta = 48.2, -48.2$	A1 [4]
		Total 12 marks

Type MS · Quadratic trig equations using an identity

Answer 8 of 11 · Question 7 part c · 4 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For any of 30 **or** 330 **or** 390 May be implied by correct answers.
	M1 A1 A1	Solves a linear equation coming from attempt at use of inverse trigonometric function to obtain one value. e.g. solves $3x - 15 = "30"$ First A1 for any correct value, second A1 for all 3 correct values and no others in the range. Ignore values outside the range.
(b)	M1	Correctly replaces the identity for tan y and attempts to deal with sin y. Allow for factorising. Condone dividing through by sin y. Minimally acceptable attempt for factorisation is $A \sin y\left(\frac{B}{\cos y}+C\right)$
	A1ft	$\sin y = 0$ and $\cos y = -\frac{B}{C}$, follow through their B and C **only.**
	A1	From sine: Both values, ignore extra values out of range, A0 for extra values in range.
	A1	From cosine: Both values, ignore extra values out of range, A0 for extra values in range. y = awrt 138.6, awrt – 138.6
	ALT	
	M1	Correctly replaces the identity for tan y and attempts to deal with tan y. Allow for factorising. Condone dividing through by tan y. Minimally acceptable attempt for factorisation is $A\tan y\,(B + C\cos y)$
	A1ft	$\tan y = 0$ and $\cos y = -\frac{B}{C}$, follow through their B and C **only.**
	A1	From tangent: Both values, ignore extra values out of range, A0 for extra values in range.
	A1	From cosine: Both values, ignore extra values out of range, A0 for extra values in range. y = awrt 138.6, awrt – 138.6
(c)	M1	Correctly uses the identity for $\cos^2\theta$ and rearranges to get a 3TQ Minimally acceptable attempt is $\pm 3\cos^2\theta \pm \cos\theta \pm 2 = 0$
	M1	Solves a 3TQ to arrive at 2 distinct values for $\cos\theta$. See general guidance.
	A1	For – 180, ignore extra values out of range, A0 for extra values in range.
	A1	Both values, ignore extra values out of range, A0 for extra values in range. θ = awrt 48.2, awrt – 48.2

Type MS · Quadratic trig equations using an identity

Answer 8 of 11 · Question 7 part c · 4 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Quadratic trig equations using an identity

Answer 9 of 11 · Question 10 part c · 5 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

Question number		
10 (a)	$(x+\frac{\pi}{3})=\frac{\pi}{3}$ or $\frac{2\pi}{3}$ or $\frac{7\pi}{3}$ $(x+\frac{\pi}{3})=\frac{\pi}{3}$ and $\frac{2\pi}{3}$ and $\frac{7\pi}{3}$ $x=0,\frac{\pi}{3},2\pi$	M1 A1 A1 (3)
(b)	$\tan\theta=-\frac{5}{3}$ $\theta = -59^\circ, -239^\circ, 121^\circ, 301^\circ$	M1 M1 A1 (3)
(c)	$1+\sin 2y-2(1-\sin^2 2y)=0$ $2\sin^2 2y+\sin 2y-1=0$ $(\sin 2y+1)(2\sin 2y-1)=0$ $\sin 2y=-1$ or $\sin 2y=\frac{1}{2}$ $2y=-90^\circ, (30^\circ), (150^\circ), -330^\circ, -210^\circ$ $y=-45^\circ, -105^\circ, -165^\circ$	M1 A1 dM1 A1 A1 (5) **[11]**

Part	Mark	Additional Guidance
(a)	M1	Any one of the three indicated angles, radians only, ignore any other angles.
	A1	For all three indicated angles, ignore other angles out of the range $\frac{\pi}{3}\le x+\frac{\pi}{3}\le\frac{7\pi}{3}$
	A1	For all three angles, ignore angles out of range, A0 if additional angles in range.
(b)	M1	For $\tan\theta=k$. $k\neq0$, $k\neq\pm1$
	M1	Any one correct value, does not need to be to the nearest degree. Allow one correct value to imply the first M1. Ignore any other angles.
	A1	For all four angles, ignore angles out of range, A0 if additional angles in range. All four angles must be given to the nearest degree.
(c)	M1	For the correct use of $1-\sin^2 2y$ in the equation on the left or right side, equation doesn't need to be = 0.
	A1	Correct 3TQ, must be = 0 and a valid attempt to solve leading to sin $2y$ =
	dM1	$\sin 2y=-1$ or $\sin 2y=\frac{1}{2}$ (allow sin2y = a and b from a valid attempt to solve their 3TQ). Allow sin2y to be x or any other variable.
	A1	For minimum of 3 of the 5 values shown, (including the ones in brackets), ignore other angles outside the range -360° ≤ $2y$ ≤ 360°. Allow sight of 270 if -90° present.
	A1	For all 3 values shown. Ignore extras out of range. **Rounding answers (where accuracy is specified in the question)** Penalise only once per question for failing to round as instructed - ie giving more digits in the answers.

Type MS · Quadratic trig equations using an identity

Answer 9 of 11 · Question 10 part c · 5 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Quadratic trig equations using an identity

Answer 10 of 11 · Question 10 part b · 3 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

Number		
10 (a)	$x = \tan^{-1}(-3) = -72$	M1
	$x = 108 \qquad x = 288$	A1 A1
		(3)
(b)	$7\sin^2\theta + \sin\theta\cos\theta = 6(\sin^2\theta + \cos^2\theta)$	M1
	$\sin^2\theta + \sin\theta\cos\theta - 6\cos^2\theta = 0$	
	$\frac{\sin^2\theta}{\cos^2\theta} + \frac{\sin\theta}{\cos\theta} - 6 = 0$	M1
	$\tan^2\theta + \tan\theta - 6 = 0$	A1 cso
		(3)
(c)	$(\tan y + 3)(\tan y - 2) = 0$	M1
	$\tan y = -3 \qquad \tan y = 2$	A1
	$y = 108, 288 \qquad y = 63, 243$	A1ft A1
		(4)
		Total 10 marks

Type MS · Quadratic trig equations using an identity

Answer 10 of 11 · Question 10 part b · 3 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Guidance
(a)	M1	For using inverse tan to obtain any correct angle $\tan^{-1}(-3) \Rightarrow x = -71.565...^{o}$ Accept awrt -72°
	A1	For **either** 108 **or** 288
	A1	For **both** 108 **and** 288
(b)	M1	Uses $\sin^2\theta + \cos^2\theta = 1$ on the given equation to obtain $7\sin^2\theta + \sin\theta\cos\theta = 6(\sin^2\theta + \cos^2\theta)$
	M1	For rearranging and dividing through by $\cos^2\theta$ with the $\frac{\sin\theta}{\cos\theta} = \tan\theta$ identity to obtain a 3TQ: $\frac{\sin^2\theta}{\cos^2\theta} + \frac{\sin\theta}{\cos\theta} - 6 = 0 \Rightarrow \left(\tan^2\theta + \tan\theta - 6 = 0\right)$
	ALT 1	
	M1	Divides the given equation through by $\cos^2\theta$ with the $\frac{\sin\theta}{\cos\theta} = \tan\theta$ identity to obtain $7\tan^2\theta + \tan\theta = \frac{6}{\cos^2\theta}$
	M1	Uses $\sin^2\theta + \cos^2\theta = 1$ to obtain $\tan^2\theta + 1 = \frac{1}{\cos^2\theta}$ and uses this result on the given equation and rearranges to achieve a 3TQ to obtain $7\tan^2\theta + \tan\theta = 6\left(1 + \tan^2\theta\right) \Rightarrow \left(\tan^2\theta + \tan\theta - 6 = 0\right)$
	A1	For obtaining the given expression $\tan^2\theta + \tan\theta - 6 = 0$ * in full. **Note: This is a show question, there must be no errors in the solution.**
	ALT 2	
	M1	Uses the identity $\frac{\sin\theta}{\cos\theta} = \tan\theta$ with $\tan^2\theta + \tan\theta - 6 = 0$ to achieve $\frac{\sin^2\theta}{\cos^2\theta} + \frac{\sin\theta}{\cos\theta} - 6 = 0$ and multiplies through by $\cos^2\theta$ to obtain $\sin^2\theta + \sin\theta\cos\theta - 6\cos^2\theta = 0$
	M1	Uses $\sin^2\theta + \cos^2\theta = 1$ to obtain $\sin^2\theta + \sin\theta\cos\theta - 6\left(1 - \sin^2\theta\right) = 0$ and rearranges to obtain $7\sin^2\theta + \sin\theta\cos\theta = 6$
	A1 cso	For obtaining the given expression $7\sin^2\theta + \sin\theta\cos\theta = 6$ in full. **Note: This is a show question, there must be no errors in the solution.**
(c)	M1	For changing $7\sin^2 y + \sin y\cos y = 6$ to $\tan^2 y + \tan y - 6 = 0$ [this step must be correct] and then attempting to solve the 3TQ by any method.
	A1	For $\tan y = -3$ and $\tan y = 2$
	A1ft	For **both** $y = 108$ **and** 288 (ft from (a)) **Do not ft angles out of range**
	A1	For **both** $y = 63$ **and** 243
Rounding errors: Penalise rounding only once in this question when first seen provided angles round to 108, 288, 63 or 243		
Extra angles: Deduct one A mark for any extra angles within range. Ignore angles outside of range.		

Type MS · Quadratic trig equations using an identity

Answer 10 of 11 · Question 10 part b · 3 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Quadratic trig equations using an identity

Answer 11 of 11 · Question 8 part c · 4 marks

2020 JAN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

Number		
8 (a)	$2x - 75 = -31, 211$ $x = 22, 143$	M1A1 A1 (3)
(b)	$2\frac{\sin y^\circ}{\cos y^\circ} + 5\sin y^\circ = 0$ $\sin y^\circ\left(\frac{2}{\cos y^\circ} + 5\right) = 0$ $\cos y^\circ = -\frac{2}{5}$ $(\sin y^\circ = 0)$ $y = 113.6^\circ$ $y = 0^\circ, 180^\circ$	M1 M1 A1 B1 (4)
(c)	$3(1 - \sin^2\theta) - 3\sin^2\theta + \sin\theta + 12 = 0$ $6\sin^2\theta - \sin\theta - 15 = 0$ $(2\sin\theta + 3)(3\sin\theta - 5) = 0$ $\sin\theta = -\frac{3}{2}$ $\sin\theta = \frac{5}{3}$ As $-1 \leq \sin\theta \leq 1$ no such values for θ exist	M1 M1 A1 B1 (4)
		[11]

Type MS · Quadratic trig equations using an identity

Answer 11 of 11 · Question 8 part c · 4 marks

2020 JAN · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For finding at least one correct value of $(2x-75)=-31^\circ$ or 211° and attempting to find one value of $x \Rightarrow x=\frac{-31+75}{2}$ or $x=\frac{211+75}{2}$
	A1	For $x=22$ or 143
	A1	For $x=22$ and 143 Extra values within range – A0 Extra values outside of the range - ignore
.(b)	M1	For using the identity $\tan y^\circ=\frac{\sin y^\circ}{\cos y^\circ}$
	M1	For factorising their expression and finding values for $\sin y^\circ$ and $\cos y^\circ$ $\sin y^\circ\left(\frac{2}{\cos y^\circ}+5\right)=0 \Rightarrow \sin y^\circ=0,\ \cos y^\circ=-\frac{2}{5} \Rightarrow y=...$
	A1	For $y=113.6$ if there are extra values within range – A0
	B1	For both $y=0$ and 180 **Both required**
	ALT	
	M1	For multiplying $\sin y\times\frac{\cos y}{\cos y} \Rightarrow \tan y\cos y \Rightarrow (2\tan y+5\tan y\cos y=0)$
	M1	For factorising the above expression and finding values for $\tan y^\circ$ and $\cos y^\circ$ $2\tan y+5\tan y\cos y=0 \Rightarrow \tan y(2+5\cos y)=0$ $\Rightarrow \tan y=0,\ \cos y=-\frac{2}{5} \Rightarrow y=...$
	A1	For $y=113.6$ Extra values within range – A0 Extra values outside of the range - ignore
	B1	For both $y=0$ and 180 **Both required**
	SC	$2\frac{\sin y}{\cos y}=-5\sin y \Rightarrow \cos y=-\frac{2}{5} \Rightarrow y=113.6$ no evidence of factorising – award M1M0A1B0 only (unless there is later recovery)
(c)	M1	For using the identity $\cos^2\theta+\sin^2\theta=1 \Rightarrow 3(1-\sin^2\theta)-3\sin^2\theta+\sin\theta+12=0$ to form a 3TQ in terms of $\sin\theta$ Minimally acceptable attempt is $6\sin^2\theta\pm\sin\theta\pm15=0$
	M1	For an attempt to solve their 3TQ (see general guidance for the definition of an attempt $6\sin^2\theta-\sin\theta-15=0 \Rightarrow (2\sin\theta+3)(3\sin\theta-5)=0 \Rightarrow \sin\theta=...,...$
	A1	$\sin\theta=-\frac{3}{2},\ \frac{5}{3}$
	B1	For the conclusion; $\lvert\sin\theta\rvert>1$ therefore no values exist for $\sin\theta$ Do not accept 'undefined' without an explanation that $\lvert\sin\theta\rvert>1$
Penalise rounding only once in this question		

Type MS · Quadratic trig equations using an identity

Answer 11 of 11 · Question 8 part c · 4 marks

2020 JAN · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks